

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

IT-833 (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
Artificial Intelligence
(Elective - III)
Time : Three Hours

Maximum Marks : 70

- Note :** i) Attempt any five questions.
 ii) All questions carry equal marks.

1. Write and explain hill climbing algorithm. What are the problem associated with this and how it can be removed?

OR

Write and explain A* algorithm. Identify the conditions at which the A* will become

- i) Breadth first search
 ii) Best first search
2. What are the characteristics of a good knowledge representation technique? Explain each of them in brief.
3. a) Elaborate on forward and backward chaining.
 b) Write short notes on:
 i) Frames
 ii) Fuzzy logic
4. a) Distinguish between supervised and unsupervised learning.
 b) Discuss back propagation algorithm for learning in multilayer neural network.
5. Solve 8 puzzle problem using best first search algorithm.

1		2		1	2	3
4	5	3	→	4	5	6
7	8	6		7	8	

6. Define Semantic Nets? Draw Semantic Nets for the following sentences -
 i) Every dog bites a mail carrier
 ii) John gave a book to marry

[2]

7. Explain in detail about Active and Passive reinforcement learning.
8. Answer any three of the following:
 - i) Explain your own criteria for computer software to be consider intelligent.
 - ii) Discuss various approaches of knowledge representation with the help of example.
 - iii) Briefly describe semantic net and partitioned semantic net.
 - iv) Why do game playing programs work from a current state to a goal state, rather than backward from a goal?
 - v) Differentiate between expert system and decision support system.
